

CinePal: Conversational Clustering

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1 Introduction

CinePal is a conversational clustering system designed for exploring movie catalogs through natural dialogue. Rather than using static operations, the system continuously partitions the database into named, inspectable clusters that the user navigates and refines throughout the conversation. Because there is **no intrinsic ground truth for cluster quality**, the user’s satisfaction serves as the explicit objective function.

The core challenge lies in inferring user **intent**, translating it into operations, and maintaining a legible structure while recovering gracefully from errors. To sustain this workflow, **label persistency** stabilizes cluster names across turns. At the same time, a content-addressed snapshot tree accommodates **evolving preferences** by enabling seamless undo, reset, and branching operations.

Ultimately, CinePal is a **build-heavy** project. Its primary contribution is the system itself, which integrates a multi-agent conversational architecture, a reproducible multi-modal data pipeline, and an automated evaluation harness.

2 System Architecture

This section describes how the system is built: the offline data pipeline that produces the embedded catalogue, the components that make up the conversational loop and the lifecycle of a single turn.

2.1 Data Pipeline

The movie catalog is constructed from The Movie Database (TMDB) through a reproducible, multi-modal ingestion pipeline. To transform raw entertainment metadata into a vector-optimized database, the pipeline processes the source data across three main stages, after which the fully embedded catalog is partitioned into three slices for downstream use: a held-out evaluation set, a main operational set, and a compact development set.

Scrape: Ingestion begins by downloading TMDB’s daily catalog export, after which the surviving titles are enriched through detailed API requests with credits, user reviews, keywords, and trailer links.

Clean & Filter: To maintain data quality and processing efficiency, the pipeline discards adult content and applies a popularity threshold before those detailed requests, retaining only the titles worth embedding.

Embed: Each movie is encoded into dense vectors across three distinct modalities. First, a **composite text** (title, synopsis, genres, cast, director, and keywords) is encoded with the *BAAI BGE-large-en-v1.5* sentence-embedding model into a 1024-dimensional, l_2 -normalized vector. Then, the aggregated **user review text** (only

top reviews) is encoded by the same model into the same 1024-dimensional space. Finally, the **visual modality** is captured by sampling evenly spaced frames from the movie’s trailer, embedding them with the *OpenCLIP ViT-H/14* vision encoder, and mean-pooling them into a single 1024-dimensional vector.

Fuse: Text and review embeddings are merged directly via a weighted average of their vectors since they share a underlying space. In contrast, the text and trailer modalities are fused at the distance level using a weighted sum of their respective cosine-distance matrices, combining their structural signals without forcing them into an artificial, unified space.

2.2 Architectural Overview

The system is a small set of single-purpose components coordinated around one orchestrator (Figure 1).

Coordinator: The **deterministic** orchestrator of the system, acting as both the session handler and the turn workflow manager. As the only writer of state and the sole component that communicates directly with the oracle, it directs the system’s execution during each turn and provides the necessary context to each downstream agent. After the Intent Agent converts the oracle message into an ordered action list, the coordinator routes that plan through the turn pipeline, deciding what to clarify, what to execute, and what state to persist. The coordinator maintains **no in-memory session state**; instead, it reloads the current snapshot and re-reads the database at every turn, ensuring its behavior depends entirely on the persisted dialogue history.

Intent Agent: The **strong-model** parser that maps the oracle’s free-form message onto an **ordered list of actions**, each carrying a target cluster, operation parameters, and a confidence score. A single message can resolve into several actions executed in order, so that later steps act on the result of earlier ones (for example, *reset, then split by genre*). Rather than operating on raw embeddings, it evaluates the message against rich contextual signals, reading the user’s input alongside the recent dialogue transcript, pending clarifier questions, and the current cluster labels and summaries. Every action is drawn from a fixed, closed vocabulary, split into **navigation** operations that reshape the clustering and **dialogue** moves that do not:

- **CLUSTER:** split a cluster (or the whole catalogue) into sub-groups, by free density, a concept axis, or a metadata partition;
- **MERGE:** combine two or more clusters into one;
- **FOCUS:** keep a single cluster and discard the rest;
- **CROSS-FILTER:** keep only the films matching a metadata predicate;
- **EXCLUDE:** drop one cluster and keep the others;
- **RESET:** return to the unclustered state;

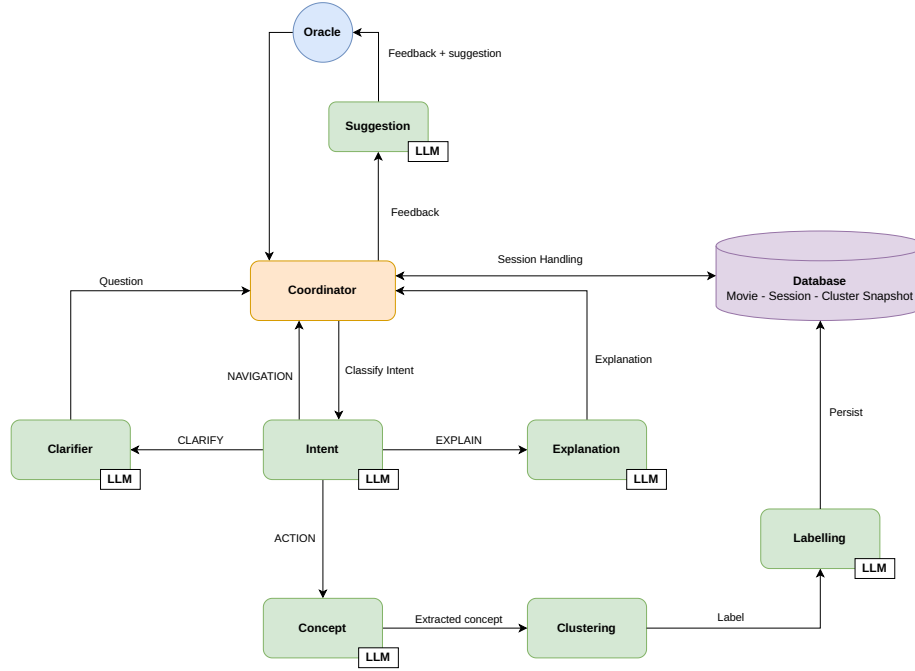


Figure 1: System components and the path of a single turn. The oracle’s message enters the deterministic Coordinator, which routes it through read-only agents.

- **UNDO**: step back to the previous snapshot;
- **EXPLAIN**: say why a given film sits in its cluster;
- **SMALL-TALK**: a casual, non-operational message.

Concept Agent: Triggered exclusively when a request is **semantic** (concept-guided) rather than a deterministic partition, this **strong-model** agent takes as input only the concept string and a declared **space** (semantic or visual). Because it is completely blind to the broader conversation, its output is a pure function of the concept itself: a **linear axis** represented as a unit vector. To build it, the agent generates k descriptive sentences for each of an engineered positive and negative pole, embeds them, and averages each set into a pole centroid; the axis then runs from the negative to the positive centroid. Each film is then scored by the dot product of its embedding with this axis, within the chosen space.

Clustering: The **deterministic** computational engine driving every semantic clustering operation. Implemented as a set of typed command classes, this engine executes a pure mathematical pipeline consisting of **UMAP** dimension reduction followed by **HDBSCAN** soft clustering. Instead of assigning rigid, hard labels, it operates exclusively on the films currently in scope and the action’s parameters to calculate and return **per-film membership probabilities** wherever the metric allows.

Labeling Agent: A **fast-model** agent responsible for naming and summarizing clusters using a strategy dictated by the operation that created them. When a cluster arises from a **deterministic partition** by a metadata attribute, the label is already implied and the agent assigns it deterministically. For **semantic or concept-guided** clusters where names cannot be mechanically derived, the

agent generates names via the LLM in a single batched call, anchoring the new labels on the previous turn’s names to prevent cosmetic churn. In both scenarios, the agent evaluates each cluster’s exemplar films alongside its inherited parent label, and the cluster’s description is always produced by an LLM call, even when the label itself is deterministic.

Clarifier Agent: A **fast-model** gate: when any action falls below the confidence threshold, it asks a single **disambiguating question** without mutating state. It sees the low-confidence action, the message behind it, and the current cluster labels.

Explanation Agent: A **strong-model** agent that explains why a given film sits in a given cluster, grounded in the cluster’s label and exemplars. It receives the cluster context (label, summary, and exemplars) and outputs a concise natural language explanation.

Suggestion Agent: A **fast-model** responder that, once actions complete, computes **deterministic signals** (a similar-cluster pair, a dominant cluster, a high noise fraction) and calls the model to propose a next step. It sees the final snapshot and the evolution trace, and may decline if it cannot identify a coherent suggestion.

2.3 Turn Handling

The operational core of the conversational loop relies on a highly synchronized, state-aware processing sequence. (Figure 2)

Initialization and Intent Parsing: The turn begins with the coordinator loading the active snapshot and labeling any remaining unlabelled clusters. It then checks whether the previous turn ended

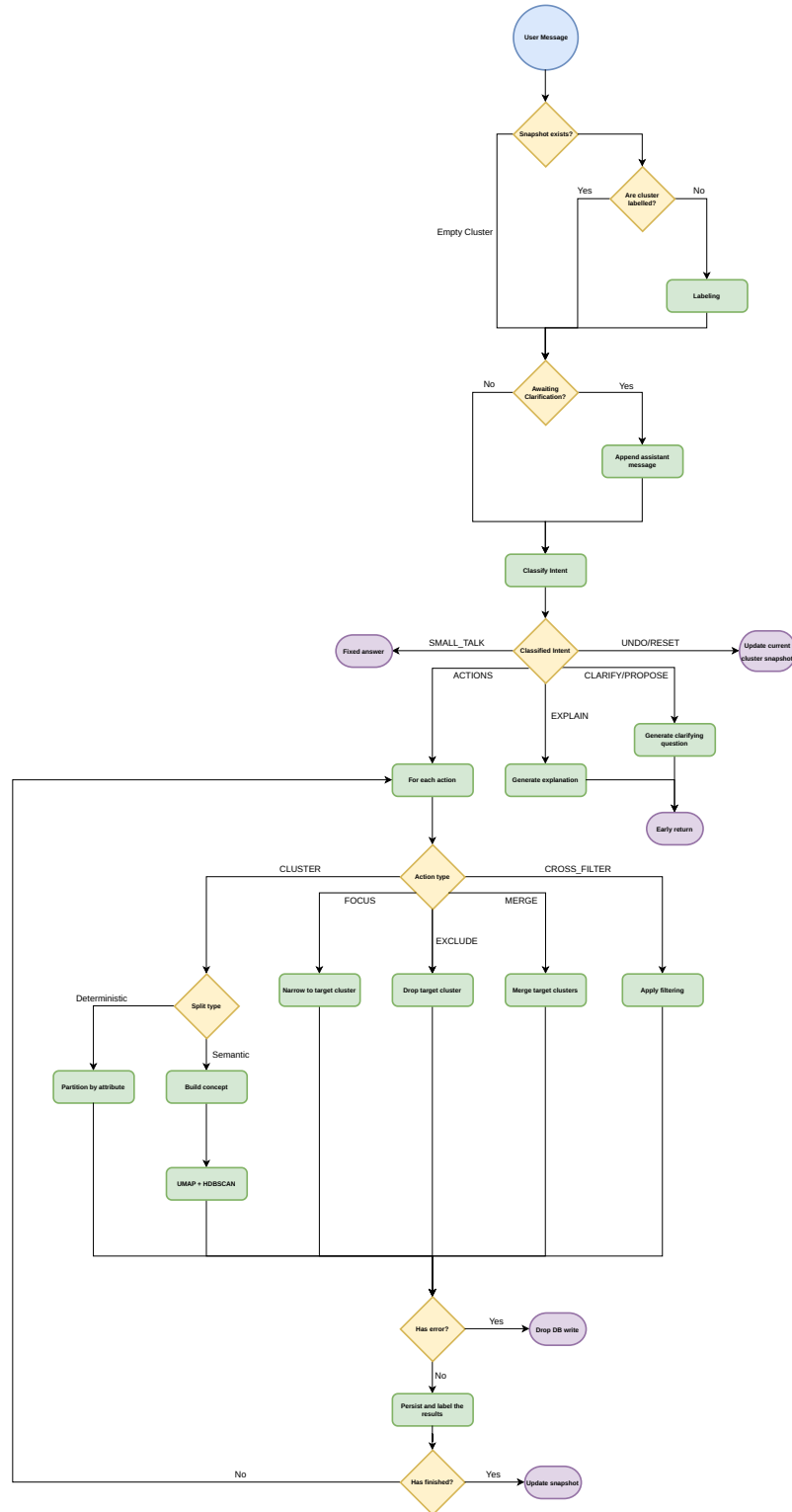


Figure 2: The per-turn lifecycle: load and label, resolve clarification context, classify intent, gate on confidence, dispatch each action to its command, and persist or drop the result.

with a clarifying question, ensuring a follow-up answer is interpreted within its proper context. With the context established, the coordinator invokes the Intent Agent to parse the message into an ordered sequence of actions.

The Confidence Gate: Before modifying any data, the system evaluates the Intent Agent’s certainty. If a state-changing action falls below the confidence threshold, the coordinator halts execution, emits a clarifying question to the oracle, and returns early without mutating state.

Execution and Persistence: If the confidence gate is cleared, the coordinator dispatches the actions sequentially to their respective deterministic commands. Each command processes the data within the state environment left by its predecessor. The coordinator only persists the new snapshot and labels upon total success; if a command throws an error, the entire write operation is dropped.

Finalization: In the final step, the coordinator may prompt the Suggestion Agent to generate recommended next steps, aggregates the resulting response fragments, and persists the final assistant message along with its operational cost.

Snapshot Graph: Rather than overwriting state data on every turn, the system records the evolution of the dialogue as a directed tree graph. Every executed operation generates a distinct, content-addressed snapshot node containing references to its parent, the specific command executed, and its operational parameters. Because nodes are content-addressed, arriving at an identical state via different paths automatically reuses the existing cached node. This branching data structure natively supports advanced navigation behaviors.

2.4 System Interfaces

The system exposes two programmable interfaces and one human-facing interface, all thin surfaces over the coordinator and the data-access layer:

- **HTTP API:** carries the full session lifecycle, creating conversations, submitting messages, reading snapshots and movies, streaming per-turn progress, and serving the evaluation data, with authentication handled entirely at this layer;
- **Web frontend:** surfaces the entire human-facing experience, from starting and replaying conversations to inspecting clusters and their member films, visualising and navigating the snapshot graph, and, for administrators, an evaluation lab that reviews run metrics, judge scores, and full transcripts;
- **MCP server:** re-exposes the same capabilities as a tools-only, stateless interface, so an external agent can drive CinePal exactly as a human client would. It deliberately stops short of the full HTTP surface, leaving out the operator and human-only concerns: authentication and per-user history, destructive hard deletes, the live progress stream, the raw UMAP projection, and the evaluation routes, none of which an autonomous client needs in order to explore the catalogue.

3 Evaluation Framework

This section describes how the system is evaluated: the overall validation design, the simulated oracle that drives the sessions, and the metrics used to score them.

3.1 Evaluation Design

Because cluster quality is defined by the oracle, the system cannot be graded against a fixed target. The evaluation instead centres on one question:

Does an intent-driven navigation loop produce more coherent clusterings of the catalogue, with less oracle effort, than other variants?

We approach it from three angles:

- **Operation choice:** given a free-form message, does the Intent agent pick the right navigation operation and the right parameters?
- **Cluster update strategy:** do concept-guided splits, soft HDBSCAN memberships, and the content-addressed snapshot tree yield more stable, more separable clusters than plain re-clustering on raw embeddings?
- **Satisfaction:** how do we assess clustering quality when there is no recommended film list, only a sequence of cluster snapshots, and how far do those assessments generalise?

To answer this, the evaluation closes a **three-party loop**: the **system under test** is driven by an **LLM-simulated oracle** that plays a user with a hidden goal, while a separate **LLM judge** scores the finished session.

3.2 Oracle Formulation

Ground truth: Each session is anchored by a ground truth: a target navigation **trajectory**, an ordered list of operations, paired with a natural-language **intent description** that paraphrases it. The oracle is shown only the intent description, in order to act as a real user would.

Persona: A persona is layered on top of the ground truth through two dials:

- **Verbosity** controls how much the oracle says;
- **Patience** controls how many misunderstandings it tolerates before giving up.

Protocol: To ensure the simulated oracle aligns with the application and evaluation’s interactive workflow, it adheres to a strict communication protocol. The oracle issues exactly one request per message, explicitly answers any clarifying questions raised by the system, and requests an explanation for why a specific film resides within its cluster exactly once per session. The session terminates organically when the oracle either achieves success, concluding with a high satisfaction rating of 4 or 5, or decides to give up after repeated system failures, yielding a low rating of 1 or 2. Finally, a strict **turn budget** serves as a hard cap to terminate the session if neither natural exit condition is met.

Reproducibility: The oracle is engineered to be fully reproducible: its output behaves as a deterministic function of the configured persona, the catalog’s ground truth, the session seed, and the active turn number. Its only source of non-determinism, the language

model’s sampling, is itself keyed on that same quadruple, so rerunning the exact configuration reproduces the same session rather than drifting.

3.3 Metrics

Evaluation runs automatically at the end of each session and produces two kinds of measurement.

Deterministic metrics. Read directly from the recorded session state, with no model involved:

- **Final cluster count:** number of clusters in the final snapshot;
- **Turns:** number of oracle turns in the session;
- **Operations:** number of navigation operations executed;
- **Clarifier trigger rate:** fraction of turns on which the clarifier fired;
- **Total cost:** cumulative LLM cost of the session;
- **Oracle rating:** the oracle’s own self-rating (1–5).

Judge dimensions. Subjective scores (1–5) assigned by the judge after reading the full transcript, the per-turn action log, and clusters.

- **Operation appropriateness:** did the system pick the right operation and parameters for each request;
- **Intent alignment:** does the final result reflect the stated goal;
- **Label accuracy:** do the cluster names and summaries match their contents;
- **Clustering coherence:** do the films within each cluster genuinely belong together;
- **Explanation quality** (conditional): is a placement explanation faithful and specific;
- **Suggestion meaningfulness** (conditional): is a volunteered suggestion relevant and well-timed;
- **Concept-axis quality** (conditional): do the axis poles form a coherent, discriminative spectrum.

Because the conversational setting lacks a rigid ground-truth trajectory, traditional supervised alignment is impossible. The oracle’s private plan is inherently flexible; its free-form messages are stochastic, meaning several distinct operational paths can validly satisfy the same underlying intent. Instead, the judge evaluates appropriateness dynamically, rating whether the system’s chosen operation and parameters constitute a logical and valid response.

Confidence Intervals. For every metric, the dashboard reports a two-sided 95% confidence interval of the mean, computed from per-session values using a t -distribution:

$$CI_{95} = \bar{x} \pm t_{0.975, n-1} \cdot \frac{s}{\sqrt{n}}, \quad (1)$$

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}.$$

where n is the session count, s is the Bessel-corrected sample standard deviation, and $t_{0.975, n-1}$ is the critical value for $n - 1$ degrees of freedom (falling back to 1.96 for $n \geq 31$). Bootstrapping is deliberately avoided: at our sample sizes ($n \approx 30$ per condition), resampling cannot recover information a small sample lacks. Non-overlapping intervals are interpreted as a practical indicator of a meaningful difference between conditions, rather than a formal hypothesis test.

4 Findings and Discussion

The main theme emerging from the evaluation is that the conversational system excels at operational fidelity and responding to user intent (in a honest way), whereas the baseline system primarily offers a tidier visual output.

Operational Appropriateness. The architecture’s primary strength is selecting the right operation and following the oracle’s trajectory. This is the only dimension with disjoint confidence intervals: operation appropriateness 3.94 ± 0.31 against the baseline’s 2.73 ± 0.35 (Table 4). It also trends higher on intent alignment (3.09 ± 0.43 vs. 2.33 ± 0.47) and on the oracle’s own rating (3.41 ± 0.49 vs. 2.82 ± 0.64). This confirms slightly success on this research angles.

Cost of the System. Going multi-agent cost per turn is \$0.0126 vs. \$0.0123, and cost per session \$0.169 vs. \$0.153 (Table 5). The overhead is absorbed because about a quarter of conversational turns are cheap non-operational moves (clarify, explain, small-talk), visible in the lower operations-per-turn rate (0.74 vs. 0.93). The usual cost objection to a modular, multi-call design therefore does not apply here, especially compared to the baseline’s.

Cluster segmentation and structure. The two systems fail in opposite directions. The baseline chronically under-segments, collapsing to two or fewer clusters in 87% of sessions (26 of 30; final cluster count 1.97 ± 0.53 , Table 5). Its higher clustering coherence (2.87 ± 0.31) and label accuracy (3.53 ± 0.36 , Table 4) are artifacts of these coarse, easy-to-name groups. Furthermore, throughout the conversation, it tends to lose films randomly, resulting in smaller clusters; this occurs even when utilizing a re-identification step for the films prior to prompting. In contrast, the conversational system usually produces a balanced handful of clusters (4.14 ± 1.67) but occasionally over-segments, exploding to 17–18 groups in four sessions. This could also happen due to the oracle approach that tends to perform more splits than necessary.

Abstract Concept Cliff. Performance splits sharply by intent type. The conversational system is an effective clusterer of well-scoped, metadata-grounded goals, with per-persona oracle ratings up to 4.67 for the oracles “genre fan” or a “rater”, but it drops to 2.0 for fuzzy semantic goals such as a satirical tone, the New Hollywood movement, or the heist sub-genre. Forced to express every concept as a discrete navigation step, the pipeline struggles to capture a holistic artistic idea that a single free-form pass can sometimes grasp.

Explanation Problem. The explanation agent grounds its rationale in the wrong film roughly half the time: of the 89 “why is film X here” turns, 43 (48%) discussed a different title, for example a question about *Don’t Look Up* answered with “Playdate” and one about *Mean Girls* with “Friends with Benefits” (failure session, Table 9). This is the lowest-scoring dimension in the study: explanation quality 1.79 ± 0.31 (Table 4).

Suggestion Problem. The judge scores suggestion meaningfulness at 2.71 ± 0.21 (Table 4), in the low-middle of the 1–5 scale, indicating that the volunteered next steps are frequently generic or redundant rather than genuinely advancing the oracle’s goal.

Labelling Problem. The labeling agent prepends each new split’s descriptor to the inherited name instead of rewriting it, so labels bloat with depth: the failure session reaches Light-hearted Romantic Comedies Asian Romantic Comedies Family Fantasy Comedies after three drill-downs. Label accuracy for the conversational system stays middling at 3.03 ± 0.44 , below the baseline’s coarser-but-cleaner 3.53 ± 0.36 (Table 4).

Oracle and Judge Divergence. Two measurement caveats stand out. First, oracle satisfaction is bimodal, clustering at the extremes (ratings 1–2 or 4–5) with few middle values, so the mean rating is a poor summary. Second, the oracle is a lenient judge of final structure: the top-rated conversational session (oracle = 5) had a judged clustering coherence of only 2, looped, and returned two groups when three were requested (Table 7). Oracle rating and judge dimensions diverge, and should be read together rather than either alone.

4.1 Future Work

Each direction below targets one of the problems surfaced above, preserving their separation.

Grounding the Explanation Agent. Addressing the *explanation problem*, the explanation agent will be bound to the exact film identifier it is asked about rather than to the cluster’s exemplars, so it can no longer justify a different title. Then it should be able to give more explanations variants, like comparing clusters, describing the cluster’s internal diversity, or listing the most distinctive attributes of the film in that cluster.

Sharpening Proactive Suggestions. Addressing the *suggestion problem*, a relevant improvement would be to provide to the suggestion agent a more structured context of the conversation, in order to make it able to identify more specific next steps that are not just generic operations.

Renaming instead of Compounding Labels. Addressing the *labelling problem*, the labeling agent should be revise to better handle deterministic label generation, also considering to delegate them to an LLM.

Bridging the Abstract Concept Cliff. To bridge the *abstract concept cliff*, future works would enrich the catalog with deep, unstructured textual signals (film scripts, production transcripts, and an expansive corpus of critical reviews) retrieved through a Retrieval-Augmented Generation (RAG) framework. Rather than re-encoding this text into the same shallow vectors, for a given concept the system would retrieve the passages most relevant to each candidate film and let an LLM score that film on the concept directly, building the axis from these judgments instead of from metadata or embedding distances. This depth is not free, however: per-film retrieval and LLM scoring is far costlier than a single embedding pass, coverage is uneven since scripts and detailed reviews are missing for many titles, and the live retrieval-and-scoring step would have to be cached to preserve the deterministic replay guarantee.

Reconciling Oracle and Judge Divergence. Addressing the *oracle and judge divergence*, a human-in-the-loop study will test how far

the agent-to-agent dynamics translate to genuine human satisfaction, and the judge will be calibrated so that its dimensions and the oracle’s rating can be trusted jointly rather than read against one another.

Scaling and Refining the Evaluation. The present study was bounded by the time and compute budget of a course project, which capped each condition at roughly thirty sessions; at that scale only operation appropriateness reaches disjoint confidence intervals, while every other difference remains a suggestive trend. A natural next step is to enlarge the campaign by an order of magnitude, broadening the pool of personas, ground truths, and random seeds so that per-persona effects no longer rest on a handful of runs and the intervals tighten enough to settle these trends, alongside refinements to the harness such as additional deterministic metrics and a broader ground-truth generator.

Dataset Diversity, Scale, and New Domains. Because the propose, partition, and refine loop is domain-agnostic, making the embedding pipeline and prompts fully configurable would let the architecture be validated beyond a single curated film catalogue. Testing on catalogues of different sizes, attribute densities, and domain characteristics, such as books, academic papers, or retail products, would surface edge cases in the clustering and labeling agents and help calibrate the cost-limit settings, which a single curated catalogue cannot reveal.

5 Ethical Considerations

Synthetic Oracle Limitations. Because all evaluations rely on an LLM-simulated oracle rather than human subjects, our findings reflect agent-to-agent dynamics rather than authentic human behavior or preferences.

Model and Embedding Bias. The language and embedding models driving the system are trained predominantly on Western, English-language data, inheriting its cultural biases.

LLM-as-a-Judge Risks. Employing an LLM to evaluate another LLM introduces severe structural risks regarding evaluation accuracy, blind spots, and hallucinations.

6 Conclusion

CinePal set out to cluster a movie catalogue not against a fixed metric but through conversation, with the oracle’s acceptance as the objective. The build-heavy contribution promised in the introduction is realized: a working multi-agent conversational architecture, a reproducible multi-modal data pipeline, and an automated oracle-and-judge evaluation harness. Driving the full system with a simulated oracle confirms that this conversational, acceptance-driven approach is viable, with its remaining weaknesses confined to localized, fixable components rather than the core design.

A Results

A.1 Experimental Setup

Dataset: The evaluation ran against a compact catalogue, held out from the production dataset, of **288 films**.

Conditions: Two conditions were compared. The **conversational** condition represents our full multi-agent system. The **baseline** is a single-prompt alternative: a single model call per turn receives the raw film list, the current cluster state, and the dialogue transcript, then emits the entire updated grouping, the chosen operation, and the conversational response in a single output. This baseline completely bypasses embeddings, density-based clustering, concept axes, clarifying loops, and dedicated labeling agents.

Models: The system, oracle, and judge run on three different model families (Table 1).

Role	Model	Family
System (strong)	GPT-4o	OpenAI
System (fast)	GPT-4o-mini	OpenAI
Oracle (simulated user)	Gemini 2.5 Flash	Google
Judge	Claude Opus 4.7	Anthropic

Table 1: Model assignment.

Parameters: All runs share the fixed configuration in Table 2.

Parameter	Value
<i>Embeddings and fusion</i>	
Text and review encoder	BAAI BGE-large-en-v1.5
Trailer encoder	OpenCLIP ViT-H/14
Embedding dimension	1024
Text-review fusion, text weight	0.6
Text-review fusion, review weight	0.4
Text-trailer fusion, text weight	0.6
Text-trailer fusion, trailer weight	0.4
<i>Clustering</i>	
HDBSCAN min. cluster size (base)	50
HDBSCAN min. cluster size (online)	10
HDBSCAN min. samples	10
UMAP components	50
UMAP neighbours	15
<i>Interaction</i>	
Intent confidence threshold	0.6
Turn budget (max. oracle turns)	15
Exemplar titles per cluster	5
<i>Cost</i>	
Cost ceiling, system (USD)	10
Cost ceiling, oracle (USD)	2
Cost ceiling, judge (USD)	2
Decoding seed, all models	42

Table 2: Fixed configuration shared across the evaluation runs.

Procedure: Each persona/seed pair initialized a session driven by the simulated oracle until termination or budget exhaustion, after which all metric and judge dimensions were logged. This process

yielded **35 conversational** sessions and **30 baseline** sessions. All reported metrics are extracted directly from the resulting database and computed using the confidence intervals from Equation 1.

A.2 Quantitative Results

We report the measured outcomes for both conditions side by side; their interpretation is deferred to Section 4. Dimensions the baseline cannot produce are marked *n/a*; the number of sessions behind each score is given in Table 3.

Score	Conversational <i>n</i>	Baseline <i>n</i>
Always-on dimensions	35	30
Explanation quality	33	<i>n/a</i>
Suggestion meaningfulness	35	<i>n/a</i>
Concept-axis quality	29	<i>n/a</i>
Oracle self-rating	29	27

Table 3: Sessions contributing to each score. The four always-on dimensions (operation appropriateness, intent alignment, label accuracy, clustering coherence) share the same counts.

Judge dimension (1–5)	Conversational	Baseline
Operation appropriateness	3.94 ± 0.31	2.73 ± 0.35
Intent alignment	3.09 ± 0.43	2.33 ± 0.47
Label accuracy	3.03 ± 0.44	3.53 ± 0.36
Clustering coherence	2.54 ± 0.29	2.87 ± 0.31
Explanation quality	1.79 ± 0.31	<i>n/a</i>
Suggestion meaningfulness	2.71 ± 0.21	<i>n/a</i>
Concept-axis quality	3.48 ± 0.35	<i>n/a</i>
Oracle self-rating	3.41 ± 0.49	2.82 ± 0.64

Table 4: Judge dimensions and oracle rating, mean $\pm$ 95% CI;

Session metric	Conversational	Baseline
Final cluster count	4.14 ± 1.67	1.97 ± 0.53
Turns per session	13.34 ± 1.08	12.40 ± 1.27
Operations per session	9.83 ± 1.16	11.50 ± 1.30
Operations per turn (pooled)	0.74	0.93
Clarifier trigger rate	0.23 ± 0.04	0.00
Cost per session (USD)	0.169 ± 0.016	0.153 ± 0.020
Cost per turn (USD, pooled)	0.0126	0.0123

Table 5: Deterministic session metrics, mean $\pm$ 95% CI; per-turn rates are pooled over the condition.

Termination outcome	Conversational	Baseline
Finished, intent fulfilled	17/35 (49%)	11/30 (37%)
Stopped, misbehaviour	12/35 (34%)	16/30 (53%)
Budget exhausted	6/35 (17%)	3/30 (10%)

Table 6: How sessions ended, by condition.

A.3 Qualitative Results

Two example sessions are shown below: one the oracle rated a success, and one it abandoned. The exchanges are condensed summaries that capture the core of each conversation, not verbatim transcripts.

A.3.1 Success: worldcinema.

Persona: verbose, patience 0.37.

Intent description: “Break the catalogue down by language to see what non-English traditions are represented, then zoom into the French-language films and see how they spread between intimate personal stories and broader societal canvases.”

Oracle: Split the entire collection by its original language.

System: Partitioned 288 movies by original language into 13 groups.

Oracle: Zoom in on the French-language cluster.

System: Focused on French-language films: 7 movies retained, all others discarded.

Oracle: Undo, then split the *entire* catalogue by language again.

System: Partitioned 7 movies by original language into 1 group.

Oracle: We’ve run into a loop, so reset everything and re-cluster.

System: Session reset. Partitioned 288 movies into 13 groups.

Oracle: Split the French films by intimate-personal vs societal narrative, into 3 groups.

System: Split into 2 sub-clusters: Intimate French, Societal French.

Judge dimension (1–5)	Score
Operation appropriateness	4
Intent alignment	4
Label accuracy	5
Clustering coherence	2
Explanation quality	2
Suggestion meaningfulness	3
Concept-axis quality	3

Table 7: Judge scores, success session.

Session metric	Value
Final clusters	2
Turns	12
Operations	8
Clarifier rate	0.09
Cost (USD)	0.15

Table 8: Session metrics, success session.

A.3.2 Failure: satire.

Persona: terse, patience 0.76.

Intent description: “Keep only comedy films, then spread them along a spectrum from warm-hearted romantic levity to biting satirical bite.”

Oracle: keep only comedy films, then group them.

System: Filter matched 74 movies. Split into 2: Asian Romantic Comedies Family Fantasy Comedies, Family Animated Adventures Family Fantasy Comedies.

Oracle: split the Asian Romantic... group into 3 groups.

System: Split into 4 sub-clusters.

Oracle: why is *Don’t Look Up* in the Family Animated... group?

System: “Playdate” is placed in this cluster due to...

Oracle: focus on the Asian-romantic group, then split it by tone into 3.

System: Split into 3 sub-clusters: Warm Romantic Comedies Asian Romantic Comedies Family Fantasy Comedies, Satirical..., Light-hearted Romantic Comedies Asian Romantic Comedies Family Fantasy Comedies.

Oracle: why is *Mean Girls* in the Warm Romantic Comedies... group?

System: “Friends with Benefits” fits this cluster because...

Oracle: we are not making progress on the original intent.

Judge dimension (1–5)	Score
Operation appropriateness	2
Intent alignment	1
Label accuracy	3
Clustering coherence	2
Explanation quality	1
Suggestion meaningfulness	2
Concept-axis quality	4

Table 9: Judge scores, failure session.

Session metric	Value
Final clusters	1
Turns	15
Operations	11
Clarifier rate	0.29
Cost (USD)	0.19

Table 10: Session metrics, failure session.